Paper List

This semester, we are reading many of the papers that the OS community has placed into the SIGOPS **Hall of Fame (https://www.sigops.org/award-hof.html)**. The SIGOPS Hall of Fame Award was instituted in 2005 to recognize the most influential Operating Systems papers that were published at least ten years in the past. We've marked those papers on our reading list that are in the Hall of Fame.

A 4-page list containing all the questions from 2nd half of the course is **here. (https://docs.google.com/document/d/1zM_qFPxlKotUtuRXpqO8_fRGwb24Eh9qQYFOPvzzxbc/edit?usp=sharing)**

Schedule (https://canvas.wisc.edu/courses/486122/pages/schedule-sp25)

General

1. **Background**
   1. Google Code University, Distributed System Design **http://www.hpcs.cs.tsukuba.ac.jp/~tatebe/lecture/h23/dsys/dsd-tutorial.html (http://www.hpcs.cs.tsukuba.ac.jp/~tatebe/lecture/h23/dsys/dsd-tutorial.html)**
2. **Communication - 739 Questions (https://canvas.wisc.edu/courses/486122/pages/rpc-questions)**
   1. **RPC:** Andrew D. Birrell and Bruce Jay Nelson **Implementing Remote Procedure Calls (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/rpc.pdf)**. ACM Trans. Comput. Syst., 2(1):39-59, 1984. **SIGOPS Hall of Fame Award**
   2. **Optional, Skim if Desired: A Critique of the Remote Procedure Call Paradigm (https://www.cs.vu.nl/~ast/Publications/Papers/euteco-1988.pdf)**, Andrew S. Tanenbaum Robbert van Renesse, 1988
3. **Failures**
   1. Jim Gray. **Why Do Computers Stop And What Can Be Done About It? (https://pages.cs.wisc.edu/~remzi/Classes/739/Fall2018/Papers/gray85-easy.pdf)** HP Labs Technical Report TR-85.7 **SIGOPS Hall of Fame Award - 739 Questions (https://canvas.wisc.edu/courses/486122/pages/why-do-computers-stop-questions)**
   2. **(https://canvas.wisc.edu/courses/486122/pages/why-do-computers-stop-questions)** Aishwarya Ganesan, Ramnatthan Alagappan, Andrea C. Arpaci-Dusseau, and Remzi H. Arpaci-Dusseau, **Redundancy Does Not Imply Fault Tolerance: Analysis of Distributed Storage Reactions to Single Errors and Corruptions, (https://research.cs.wisc.edu/wind/Publications/fast17-ganesan.pdf)** *Proceedings of the 15th USENIX Conference on File and Storage Technologies (FAST '17).* - **739 Questions (https://canvas.wisc.edu/courses/486122/pages/redundancy-does-not-imply-fault-tolerance)**

Client-Server File Systems

1. NFS - 739 Questions (https://canvas.wisc.edu/courses/486122/pages/nfs-questions)
   1. **State:** The Role of Distributed State (https://canvas.wisc.edu/courses/486122/files/50323222?wrap=1) (https://canvas.wisc.edu/courses/486122/files/50323222/download?download_frd=1) **(skip later sections on Sprite)**
      John Ousterhout, 1991
   2. **NFSv2 Background:** Design and Implementation of the Sun Network Filesystem (http://www.cs.wisc.edu/~dusseau/Classes/CS736/Papers/nfs.pdf) **(Focus on material before Implementation)**
      Sandberg, R., Goldberg, D., Kleiman, S., Walsh, D., and Lyon, B., Proceedings of the Summer 1985 USENIX Conference, Portland OR, June 1985, pp. 119-130.
   3. **NFSv3:** NFS Version 3: Design and Implementation (https://www.usenix.org/publications/library/proceedings/bos94/full_papers/pawlowski.ps) **or** pdf version (https://cdn-uploads.piazza.com/paste/lzvknjt8bz21ge/6885e063fe6402d99b917c251462bbf72ec5e2ced8121e3f342661a6957fc167/nfs3.pdf) **(Focus on material before Section 5: Performance)**
      Brian Pawlowski and Chet Juszczak and Peter Staubach and Carl Smith and Diane Lebel and Dave Hitz, In USENIX Summer 1994
   4. **NFSv4:** The NFS Version 4 Protocol (http://www.sane.nl/events/sane2000/papers/pawlowski.pdf) **(Focus on material before Section 10: Attributes)**
      Brian Pawlowski and Spencer Shepler and Carl Beame and Brent Callaghan and Michael Eisler and David Noveck and David Robinson and Robert Thurlow, Proceedings of the 2nd international system administration and networking conference (SANE2000)
2. **AFS + Coda -** 739 Questions (https://canvas.wisc.edu/courses/486122/pages/afs-and-coda-questions)
   - **AFS Background:** Scale and Performance in a Distributed File System (http://www.cs.wisc.edu/~dusseau/Classes/CS736/Papers/afs.pdf)
     Howard, J.H., Kazar, M.L., Menees, S.G., Nichols, D.A., Satyanarayanan, M., Sidebotham, R.N., and West, M.J. ACM Transactions on Computer Systems, Vol. 6, No. 1, February 1988, pp. 51-81. **SIGOPS Hall of Fame Award**
   - **Coda:** Disconnected Operation in the Coda File System (https://dl.acm.org/doi/pdf/10.1145/121132.121166)
     James J. Kistler, M. Satyanarayanan, 13th Symposium on Operating Systems Principles, Asilomar, California, pp. 213-225. October 1991. **SIGOPS Hall of Fame Award**
3. **LBFS -** 739 Questions (https://canvas.wisc.edu/courses/486122/pages/lbfs-questions)
   - Athicha Muthitacharoen, Benjie Chen (MIT), David Mazieres (NYU), A Low-Bandwidth Network File System (http://www.sosp.org/2001/papers/mazieres.pdf) **,** SOSP'01
4. **Basic Techniques: Leases**

- (Skim) Cary G. Gray and David R. Cheriton, **Leases: An Efficient Fault-Tolerant Mechanism for Distributed File Cache Consistency (http://portal.acm.org/citation.cfm?id=74850.74870)**, Proceedings of the Twelfth ACM Symposium on Operating Systems Principles (SOSP), December 1989, Litchfield Park, AZ, USA. **SIGOPS Hall of Fame Award**

Ordering

- 739 Questions **(https://canvas.wisc.edu/courses/486122/pages/logical-clocks-and-distributed-snapshots)**

1. **Clocks:** L. Lamport. **Time, Clocks, and the Ordering of Events in a Distributed System (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/lamport-clocks.pdf)**. Communications of the ACM, July 1978, pages 558-564. **SIGOPS Hall of Fame Award**
2. **Global Snapshots:** M. Chandy and L. Lamport. **Distributed Snapshots: Determining Global States of Distributed Systems (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/snapshots.pdf)**. ACM Trans. Comput. Syst., 3(1):63-75, 1985. **SIGOPS Hall of Fame Award**

Fault-Tolerance

1. **State Machine Replication:**
   Fred B. Schneider. 1990. Implementing fault-tolerant services using the state machine approach: a tutorial. ACM Comput. Surv. 22, 4 (Dec. 1990), 299–319. **https://doi.org/10.1145/98163.98167 (https://doi.org/10.1145/98163.98167)** **SIGOPS Hall of Fame Award**
2. **Primary Backup and HA-NFS: -** 739 Questions **(https://canvas.wisc.edu/courses/486122/pages/primary-backup-and-ha-nfs-questions)**
   "A Highly Available Network File Server," (https://pages.cs.wisc.edu/~remzi/Classes/739/Fall2018/Papers/bhide91highly.pdf) A.K. Bhide and E.N. Elnozahy and S.P. Morgan. *In Proceedings of the USENIX Winter Conference 1991, pp. 199-205, Jan 1991.* ***(Scanned copy with Figures!) (https://canvas.wisc.edu/courses/486122/files/50323018?wrap=1) (https://canvas.wisc.edu/courses/486122/files/50323018/download?download_frd=1)***
3. *Remus: High Availability via Asynchronous Virtual Machine Replication (Skim)* ***https://www.usenix.org/legacy/event/nsdi08/tech/full_papers/cully/cully.pdf (https://www.usenix.org/legacy/event/nsdi08/tech/full_papers/cully/cully.pdf)***
4. **Chain Replication: -** 739 Questions **(https://canvas.wisc.edu/courses/486122/pages/chain-questions)**
   Robbert van Renesse and Fred B. Schneider. **Chain Replication for Supporting High Throughput and Availability. (https://www.cs.cornell.edu/home/rvr/papers/OSDI04.pdf)** OSDI 2004.

Serverless

1. **Making Serverless Pay-For-Use a Reality with Leopard, Tingjia Cao (https://tingjia-0v0.github.io/) , Andrea C. Arpaci-Dusseau (http://www.cs.wisc.edu/~dusseau/) , Remzi H. Arpaci-Dusseau (http://www.cs.wisc.edu/~remzi/) , Tyler Caraza-Harter (https://tyler.caraza-harter.com/)**
The 22nd USENIX Symposium on Networked Systems Design and Implementation (NSDI '25), Available as: PDF (https://research.cs.wisc.edu/adsl/Publications/nsdi25-cao.pdf)

Consensus

1. **Paxos: -** 739 Questions (https://canvas.wisc.edu/courses/486122/pages/paxos-questions)
Leslie Lamport. **Paxos Made Simple. (https://www.microsoft.com/en-us/research/uploads/prod/2016/12/paxos-simple-Copy.pdf)** ACM SIGACT News, 2001.
2. **Raft: -** 739 Questions (https://canvas.wisc.edu/courses/486122/pages/raft) (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/raft.pdf)
Diego On (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/raft.pdf) garo and John Ousterhout. (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/raft.pdf) **In Search of an Understandable Consensus Algorithm (Extended Version) (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/raft.pdf)** . In 2014 USENIX Annual Technical Conference (ATC 14), 2014.
3. **Locks and Others -** 739 questions (https://canvas.wisc.edu/courses/486122/pages/chubby-questions)
   1. **Chubby:** Mike Burrows. **The Chubby Lock Service for Loosely-Coupled Distributed Systems (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/chubby-osdi06.pdf)** . OSDI 2006. **SIGOPS Hall of Fame Award**
   2. (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/Hunt.pdf) **Zookeeper:** Patrick Hunt, Mahadev Konar, Flavio Paiva Junqueira, and Benjamin Reed. **ZooKeeper: Wait-free Coordination for Internet-scale Systems. (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/Hunt.pdf)** In USENIX annual technical conference (ATC), 2010.
4. **Optimized Consensus -** 739 Questions (https://canvas.wisc.edu/courses/486122/pages/epaxos-questions)
   1. **EPaxos:** Iulian Moraru, David G. Andersen, and Michael Kaminsky. **There is more consensus in egalitarian parliaments. (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/epaxos.pdf)** In Proceedings of the Twenty-Fourth ACM Symposium on Operating Systems Principles (SOSP), 2013.
   2. Optional: **Tolerating Slowdowns in Replicated State Machines using Copilots (https://www.usenix.org/conference/osdi20/presentation/ngo)**
   Khiem Ngo, Siddhartha Sen, Wyatt Lloyd, OSDI 2020

Consistency Models

1. **Overview: (NOTE: Next time move this early in semester!) -** 739 Questions (https://canvas.wisc.edu/courses/486122/pages/consistency-questions)

   1. **Baseball:** Douglas Terry. [(http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/baseball-13.pdf) Replicated data consistency explained through baseball (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/baseball-13.pdf)]. Communications of the ACM, Dec 2013.
2. **Bayou: -** 739 Questions (https://canvas.wisc.edu/courses/486122/pages/bayou-questions)
Terry et al. Managing update conflicts in Bayou, a weakly connected replicated storage system. SOSP, 1995 (https://pages.cs.wisc.edu/~remzi/Classes/739/Fall2016/Papers/terry95managing.pdf). **SIGOPS Hall of Fame Award**
3. **Existential: -** 739 Questions (https://canvas.wisc.edu/courses/486122/pages/existential-consistency-questions)
Lu et al. Existential consistency: measuring and understanding consistency at Facebook. (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/240-lu.pdf) In Proceedings of the 25th Symposium on Operating Systems Principles (SOSP), 2015.
4. **Linearizability: -** 739 Questions (https://canvas.wisc.edu/courses/486122/pages/rifl-questions)
Collin Lee, Seo Jin Park, Ankita Kejriwal, Satoshi Matsushita, and John Ousterhout. (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/rifl.pdf) Implementing Linearizability at Large Scale and Low Latency (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/rifl.pdf). In Proceedings of the 25th Symposium on Operating Systems Principles (SOSP), 2015.
5. **Occult: -** 739 Questions (https://canvas.wisc.edu/courses/486122/pages/occult-questions) **--** NSDI'17 Talk (https://www.usenix.org/conference/nsdi17/technical-sessions/presentation/mehdi) **(20 minutes)**
Syed Akbar Mehdi, Cody Littley, Natacha Crooks, Lorenzo Alvisi, Nathan Bronson, and Wyatt Lloyd. I can't believe it's not causal! scalable causal consistency with no slowdown cascades (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/occult.pdf). In 14th USENIX Symposium on Networked Systems Design and Implementation (NSDI 17), 2017.
6. **Nilext: -** 739 Questions (https://canvas.wisc.edu/courses/486122/pages/nilext-questions)
Exploiting Nil-Externality for Fast Replicated Storage (https://research.cs.wisc.edu/wind/Publications/nilext.pdf)
Aishwarya Ganesan, Ramnatthan Alagappan, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau
Proceedings of the 28th ACM Symposium on Operating Systems Principles (SOSP '21)

Distributed Storage and Databases

1. **GFS: -** 739 Questions (https://canvas.wisc.edu/courses/486122/pages/gfs-discussion-questions)
   1. S. Ghemawat, H. Gobioff, and S.-T. Leung. **The Google File System (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/gfs.pdf)**. SIGOPS Oper. Syst. Rev., 37(5):29-43, 2003. **SIGOPS Hall of Fame Award**
2. **Dynamo: -** 739 Questions (https://canvas.wisc.edu/courses/486122/pages/dynamo-discussion-questions)

1. Giuseppe DeCandia, Deniz Hastorun, Madan Jampani, Gunavardhan Kakulapati, Avinash Lakshman, Alex Pilchin, Swami Sivasubramanian, Peter Vosshall and Werner Vogels. **Dynamo: Amazon's Highly Available Key-Value Store. (https://www.allthingsdistributed.com/files/amazon-dynamo-sosp2007.pdf)** Proceedings of the 21st ACM Symposium on Operating Systems Principles, Stevenson, WA, October 2007. **SIGOPS Hall of Fame Award**
2. **Optional:** **Amazon DynamoDB: A Scalable, Predictably Performant, and Fully Managed NoSQL Database Service, (https://www.usenix.org/conference/atc22/presentation/elhemali)** Mostafa Elhemali, Niall Gallagher, Nicholas Gordon, Joseph Idziorek, Richard Krog, Colin Lazier, Erben Mo, Akhilesh Mritunjai, Somu Perianayagam, Tim Rath, Swami Sivasubramanian, James Christopher Sorenson III, Sroaj Sosothikul, Doug Terry, Akshat Vig, *Amazon Web Services, 2022 USENIX Annual Technical Conference (USENIX ATC 22)*

3. **Bigtable: -** **739 Questions (https://canvas.wisc.edu/courses/486122/pages/bigtable-questions)** Fay Chang, Jeffrey Dean, Sanjay Ghemawat, Wilson C. Hsieh, Deborah A. Wallach, Mike Burrows, Tushar Chandra, Andrew Fikes, and Robert E. Gruber. **Bigtable: A Distributed Storage System for Structured Data (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/bigtable.pdf)** . In Seventh Symposium on Operating System Design and Implementation (OSDI), Seattle, WA, November, 2006. **SIGOPS Hall of Fame Award**
4. **Spanner: -** **739 Questions (https://canvas.wisc.edu/courses/486122/pages/spanner)** James C. Corbett, Jeffrey Dean, Michael Epstein, Andrew Fikes, Christopher Frost, J. J. Furman, Sanjay Ghemawat, Andrey Gubarev, Christopher Heiser, Peter Hochschild, Wilson Hsieh, Sebastian Kanthak, Eugene Kogan, Hongyi Li, Alexander Lloyd, Sergey Melnik, David Mwaura, David Nagle, Sean Quinlan, Rajesh Rao, Lindsay Rolig, Yasushi Saito, Michal Szymaniak, Christopher Taylor, Ruth Wang, and Dale Woodford. 2013. Spanner: Google's Globally Distributed Database. ACM Trans. Comput. Syst. 31, 3, Article 8 (August 2013), 22 pages. **https://doi.org/10.1145/2491245 (https://doi.org/10.1145/2491245)**

Byzantine

1. **PBFT:** Miguel Castro and Barbara Liskov. **Practical Byzantine Fault Tolerance (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/osdi99.pdf)** . OSDI, 1999.

2. **Byzantine Generals [OPTIONAL]:** L. Lamport, R. Shostak, and M. Pease. **The Byzantine Generals Problem (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/byzantine-generals.pdf)** . ACM Transactions on Programming Languages and Systems, July 1982, pages 382-401.

Other Relevant Papers (may or may not be covered)

- **Causal:**
  SOSP'11 Talk (30 minutes) (https://www.youtube.com/watch?v=jh9P1moDpAc)

[(https://www.youtube.com/watch?v=jh9P1moDpAc)](https://www.youtube.com/watch?v=jh9P1moDpAc)

Wyatt Lloyd, Michael J. Freedman, Michael Kaminsky, and David G. Andersen. 2011. Don't settle for eventual: scalable causal consistency for wide-area storage with COPS. In Proceedings of the Twenty-Third ACM Symposium on Operating Systems Principles (SOSP '11). Association for Computing Machinery, New York, NY, USA, 401–416. **https://doi.org/10.1145/2043556.2043593 (https://doi.org/10.1145/2043556.2043593)**

- Daniel J. Scales, Mike Nelson, and Ganesh Venkitachalam **The Design of aPractical System for Fault-Tolerant Virtual Machines (https://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/vm-ft.pdf)** . OSR 2010.
- **Highly Available Transactions: Virtues and Limitations, (http://www.vldb.org/pvldb/vol7/p181-bailis.pdf)** Peter Bailis, Aaron Davidson, Alan Fekete† , Ali Ghodsi, Joseph M. Hellerstein, Ion Stoica, VLDB 2013
- **Chord: Chord: A Scalable Peer-to-peer Lookup Protocol for Internet Applications (https://pdos.csail.mit.edu/papers/ton:chord/paper-ton.pdf) , 2001. SIGOPS Hall of Fame Award**
- **SpecPaxos:** Dan RK Ports, Jialin Li, Vincent Liu, Naveen Kr Sharma, and Arvind Krishnamurthy. **Designing distributed systems using approximate synchrony in data center networks (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/specpaxos-nsdi15.pdf)** . In 12th USENIX Symposium on Networked Systems Design and Implementation (NSDI 15). 2015.
- **Two-Phase Commit: Butler Lampson, and David B. Lomet. (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/lampson-2pc.pdf) A new presumed commit optimization for two phase commit (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/lampson-2pc.pdf)** . VLDB. Vol. 93. 1993.
- **Agility and Performance in Elastic Distributed Storage (https://www.pdl.cmu.edu/PDL-FTP/CloudComputing/a16-xu.pdf)** LIANGHONG XU, JAMES CIPAR, ELIE KREVAT, ALEXEY TUMANOV, and NITIN GUPTA, Carnegie Mellon University MICHAEL A. KOZUCH, Intel Labs GREGORY R. GANGER, Carnegie Mellon University
- Anuj Kalia, Michael Kaminsky, and David Andersen. **Datacenter RPCs can be General and Fast (http://pages.cs.wisc.edu/~ra/Classes/739-sp20/papers/nsdi19-kalia.pdf)** .
- **Maurice P. Herlihy and Jeannette M. Wing. 1990. Linearizability: a correctness condition for concurrent objects. ACM Trans. Program. Lang. Syst. 12, 3 (July 1990), 463–492. https://doi.org/10.1145/78969.78972 (https://doi.org/10.1145/78969.78972)**
- **Lamport, "How to Make a Multiprocessor Computer That Correctly Executes**

Multiprocess Programs," (https://ieeexplore.ieee.org/document/1675439) in *IEEE Transactions on Computers*, vol. C-28, no. 9, pp. 690-691, Sept. 1979, doi: 10.1109/TC.1979.1675439.

Other Relevant SIGOPS Hall of Fame Papers (probably not covered)

- Andrew D. Birrell, Roy Levin, Roger M. Needham, and Michael D. Schroeder, **Grapevine: An Exercise in Distributed Computing (http://portal.acm.org/citation.cfm?id=358468.358487)**, Proceedings of the Eighth ACM Symposium on Operating Systems Principles (SOSP), December 1981, Pacific Grove CA, USA.
- Brian M. Oki, Barbara H. Liskov. **Viewstamped Replication: A New Primary Copy Method to Support Highly-Available Distributed Systems (http://dl.acm.org/citation.cfm?id=62549)** Proceedings of the Seventh Annual ACM Symposium on Principles of Distributed Computing (PODC 1988), Toronto, ON, Canada, Aug 1988, pp 8–17.